

Module 6: Metabolism - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What is the main role of ATP in a cell?

- A. To store long-term genetic information
- B. To carry and deliver usable energy for cellular work
- C. To act as a structural membrane
- D. To speed up reactions as an enzyme
- Answer: B
- Why: ATP is the cell's energy currency: it couples energy-releasing reactions to energy-requiring ones.

2. How does an enzyme speed up a chemical reaction?

- A. By lowering the activation energy needed to start it
- B. By adding heat to the whole cell
- C. By becoming a permanent part of the product
- D. By raising the activation energy
- Answer: A
- Why: Enzymes lower activation energy. They are catalysts and are not used up in the reaction.

3. Raising temperature far above normal can stop an enzyme from working because:

- A. all substrates suddenly disappear
- B. ATP levels increase
- C. the reaction runs backward forever
- D. the enzyme denatures and loses its active-site shape
- Answer: D
- Why: Excess heat denatures the enzyme. Without its shape, the substrate no longer fits the active site.

4. How are photosynthesis and cellular respiration related?

- A. They are exactly the same reaction.
- B. Neither one uses energy.
- C. Photosynthesis stores energy in sugars, and respiration releases it.
- D. Both occur only in the dark.
- Answer: C
- Why: Photosynthesis builds sugars using light energy, and respiration harvests that stored energy. The two processes are complementary.